PROJECT RESEARCH DESIGN

Computational Methodology

Algorithms and techniques for each phase of the motor insurance pricing study

Project: Improving Motor Insurance Pricing Through Joint Modelling of Claim Frequency and Claim Severity

Dataset: freMTPL2 French Motor Third-Party Liability

> This document turns the research questions into a practical computational plan. It specifies the recommended core algorithms, the role of each technique, the target scale, and how comparisons should be validated. Optional extensions are marked separately so the project remains focused and defensible.

Primary comparison: actuarial GLM baselines versus exposure-aware gradient boosting, plus direct Tweedie and frequency-severity pure premiums. A neural network is a planned extension if it can be validated on the same leakage-safe splits.

Method at a glance

Phase	Core algorithms / techniques	Primary output

1. Data audit	Schema/range checks, exact policy-ID parsing, duplicate/missing-value audit, outer-join reconciliation	Documented analysis population and cleaning log
2. EDA and feature preparation	Exposure-weighted summaries, distribution/relationship plots, one-hot encoding, train-fitted preprocessing	Feature/target definitions and EDA findings
3. Experimental design	Policy-level train/validation/test split; grouped cross-validation for tuning	Leakage-safe, reproducible evaluation sets
4. Frequency	Poisson GLM with log-exposure offset; Poisson gradient boosting; optional Poisson neural network	Claims per policy-year / expected claim count
5. Severity	Gamma log-link GLM; Gamma-loss gradient boosting; optional lognormal/NN benchmark	Mean claim amount conditional on a claim
6. Pure premium	Frequency × severity; Tweedie GLM and optionally Tweedie boosting	Expected claim cost per unit exposure
7. Evaluation	Distribution deviance, exposure-weighted calibration/lift by risk decile, aggregate loss, Gini/Lorenz	Comparable holdout results
8. Explanation and group checks	SHAP, group calibration/O:E and loss metrics for age and region	Interpretable findings and limitations

1. Define the targets and units first

The frequency file is at policy level; the severity file is at claim level. Do not mix their rows or treat an unobserved severity as zero. Let `N_i` be policy i's claim count, `e_i` its exposure, and `S_ij` the amount of claim j.

Model target	Definition	Model prediction
Frequency rate	`N_i / e_i`	Expected claims per policy-year
Expected count	Count target `N_i` with offset `log(e_i)`	Expected claims during observed exposure
Severity	Claim-level amount `S_ij > 0`	Expected cost per claim conditional on a claim
Pure premium rate	Policy aggregate cost divided by exposure	Expected claim cost per unit of exposure

For a policy, the frequency-severity prediction is `predicted frequency rate × predicted mean severity`. Expected total loss over its exposure is pure-premium rate times exposure. State the monetary unit and confirm whether amounts are nominal.

2. Phase 1 — Data audit and reconciliation

Required techniques: exact ID parsing; schema and type checks; missingness/duplicate audits; valid-range checks; full outer join of claim counts and policy IDs; severity aggregation by policy; explicit discrepancy classification.

- Read IDs as strings, parse numeric/scientific-notation IDs exactly, and verify integer-valued IDs before conversion.
- Validate positive exposure, nonnegative claim count, positive claim amount, and expected categorical levels.
- Aggregate claim rows by policy and compare severity-row counts with `ClaimNb`; identify severity IDs missing from frequency.
- Log every exclusion, correction, and assumption; retain raw files unchanged.

> **Dataset-specific gate:** the supplied files have 678,013 frequency policies and 26,639 severity rows. The prior audit found 195 severity rows across six policy IDs absent from frequency, plus a matched-policy count discrepancy. Do not silently drop, impute, or allocate these records. Define the valid modeling population using the dataset documentation and a written rule; report sensitivity to that rule if material.

Do not use: random row deletion to force the totals to match, filling missing severity with zero for a claimant, or joining claim rows to a policy table without checking join cardinality.

3. Phase 2 — EDA and feature preparation

Required techniques: exposure-weighted portfolio summaries, distribution visualization, subgroup claim-rate analysis, consistent categorical encoding, and train-only preprocessing.

- Summarize zero-claim share, total claims, total exposure, overall claims per policy-year, and claim-amount distribution.
- For categories, use claim rate = total claims / total exposure, not raw claim count divided by policy count. Show exposure and counts alongside rates.
- Inspect numerical nonlinearities and interactions, especially driver age, vehicle power/age, bonus-malus, and density.
- Use one-hot encoding for GLMs and tree models; scale continuous predictors for a neural network only. Fit imputers/encoders/scalers using training data within a pipeline.
- Do not oversample zero-claim policies as if this were ordinary binary classification. The target is a count with exposure, not claim/no-claim classification.

4. Phase 3 — Leakage-safe experimental design

Required techniques: fixed-seed policy-level holdout split; grouped cross-validation for hyperparameter tuning; consistent partitions for both files.

1. Split unique policy IDs into training, validation, and final test partitions (for example 64% / 16% / 20%).
2. Assign every claim row to the partition of its policy ID after reconciliation. No policy's claims may appear in more than one partition.
3. Use grouped k-fold cross-validation by policy ID inside development data if tuning is substantial.
4. Fit all preprocessing, feature selection, and target transformations on training folds only.
5. Use validation data to choose models and hyperparameters. Keep the final test set untouched until the design is fixed.

If a reliable policy-period variable becomes available, consider temporal validation as a separate robustness study. Do not invent a time order from the policy ID.

5. Phase 4 — Claim frequency algorithms

Baseline: Poisson generalized linear model

Fit a Poisson GLM with log link and offset `log(exposure)`:

```
N_i ~ Poisson(mu_i)
log(mu_i) = log(e_i) + beta_0 + beta_1*x_i1 + ... + beta_p*x_ip
frequency rate_i = mu_i / e_i
```

Use numerical risk factors and categorical factors (brand, fuel, area, region). Inspect convergence, residual/deviance diagnostics, coefficient stability, and overdispersion. This is the primary actuarial benchmark.

Nonlinear comparison: Poisson gradient boosting

Fit XGBoost with a Poisson objective (or LightGBM Poisson objective) to claim count with an exposure offset where supported, or to claim rate with exposure sample weights. The latter targets rate while weighting longer exposures more strongly. State the implementation explicitly; do not compare unweighted rates as though exposure were equal.

Tune tree depth/leaves, learning rate, number of boosting rounds, row/column subsampling, and regularization using validation or grouped CV. Use early stopping only against a development validation fold.

Planned extension: feed-forward neural network

Use a modest multilayer perceptron with embedded/one-hot categorical variables, standardized numeric features, and a Poisson likelihood. Model `log(mu_i) = log(e_i) + f_theta(x_i)`, so exposure enters as a fixed offset. Use early stopping and regularization. Include only if it can be evaluated under the same split and metric; it is not a substitute for the GLM benchmark.

Optional diagnostic: if Poisson dispersion diagnostics indicate substantial overdispersion, fit a Negative Binomial GLM as an additional statistical sensitivity model. Keep the Poisson GLM as the prespecified baseline for the research question.

6. Phase 5 — Claim severity algorithms

Train on positive claim records only, retaining the validated policy features for each claim. Split by policy ID to prevent claim rows from one policy crossing data partitions.

Baseline: Gamma GLM with log link

Use a Gamma GLM for positive, right-skewed claim amounts. Check distributional fit, residuals, and extreme-loss influence. This estimates mean severity conditional on a claim.

Nonlinear comparison: Gamma-loss boosting

Use a Gamma objective in XGBoost or LightGBM for claim amounts. Tune capacity and regularization on development data. Compare claim-level predictions on the same held-out policies. If the chosen implementation lacks a reliable Gamma objective, use a validated alternative (e.g., lognormal regression with bias correction) and clearly name it.

Optional extension: compare a lognormal severity model or a neural network with a positive-valued distributional output. Do not log-transform the target and simply exponentiate predictions without correcting retransformation bias.

Report severity metrics on claim records. The severity model is conditional on a claim and must not be evaluated by treating zero-claim policies as zero-severity observations.

7. Phase 6 — Pure-premium algorithms

Model A: Frequency-severity (two-part) estimate

Multiply the predicted annual claim frequency rate by predicted mean claim severity. This directly tests whether jointly using the two components yields a useful expected claim-cost rate. Check component calibration as well as final pure-premium calibration.

Model B: Tweedie GLM

Construct policy aggregate claim cost from the reconciled claim file; define pure-premium rate as aggregate cost / exposure. Fit a Tweedie GLM with log link, exposure handled as an offset or exposure weights on the rate target. Select variance power $1 < p < 2$ using development validation. This compound Poisson-Gamma model supports a point mass at zero and continuous positive costs.

Model C: Tweedie boosting

Optionally fit XGBoost/LightGBM Tweedie regression with an appropriate power parameter and exposure handling. Include it only if the installed implementation supports the required objective and its behavior is verified. This is the nonlinear counterpart to the Tweedie GLM.

> **Required comparison for the project question:** compare a direct Tweedie pure-premium estimate against the frequency × severity estimate on the same held-out policies, using the same accepted claim records, exposure definition, and evaluation metrics.

8. Phase 7 — Tuning and model selection

Required techniques: prespecified candidate models; random search or a small structured grid; grouped validation; identical candidate features and split; early stopping for boosting/NN; final refit on development data.

- Start with a small, justified search space. Avoid an excessively large optimization exercise for every algorithm.
- Use Poisson deviance for frequency tuning, Gamma/deviance or likelihood metrics for severity, and Tweedie deviance for pure-premium tuning.
- Use cross-validation or a dedicated validation set, never the final test set, to select hyperparameters or Tweedie power.
- Record random seed, folds, search space, best parameters, software versions, and stopping criteria.
- Refit the selected configuration on training plus validation data before final test evaluation.

9. Phase 8 — Evaluation, calibration, and ranking

Target	Primary metrics	Essential calibration check
Frequency	Exposure-weighted Poisson deviance; observed/predicted claim totals	Observed vs predicted claims per exposure by prediction decile
Severity	Gamma deviance or mean log-likelihood; MAE as an interpretable supplement	Mean observed vs predicted claim amount by risk band
Pure premium	Tweedie deviance (state power); exposure-weighted MAE; aggregate loss ratio	Observed vs predicted total cost by risk decile

- For frequency, evaluate predicted rates against observed rates with exposure weights, or evaluate counts with the appropriate expected-count formulation.
- For pure premium, evaluate only against the policy-level cost target built under the finalized reconciliation rule.
- Construct deciles using predictions from the evaluation set; report exposure and observed claim/cost volume in each bin.
- Report Lorenz/Gini ranking with the exact formula and weighting convention. Define whether ranking uses predicted cost rate or expected cost at observed exposure.
- Use paired policy-level bootstrap intervals for key metric differences if feasible; preserve policy clusters when resampling.

Do not rely on accuracy, ROC-AUC, or class balancing for the primary frequency task: those are classification concepts and do not answer whether a count/rate pricing model is accurate.

10. Phase 9 — Explainability and group analysis

SHAP explanation

Use TreeSHAP for the selected boosted model to report global importance, effect direction, and selected nonlinear interactions. Compare with GLM coefficients cautiously; coefficients and SHAP values have different meanings. State the sample, feature encoding, and prediction scale used.

Age and region analysis

For age bands and regions, report group exposure, observed and predicted claim frequency/cost, observed-to-expected ratio, calibration error, and uncertainty. Compare predictive performance across groups; do not infer causal effects or claim fairness from a single metric.

Define age bands before inspecting final test results or use continuous-age calibration plus a transparent banding rule. Show small group sizes and limitations. Discuss that rating-variable relevance and fairness are separate policy/legal questions; this analysis is diagnostic, not legal advice.

11. Phase 10 — Final implementation and reporting

1. Freeze the reconciliation rule, targets, split IDs, metrics, and candidate model definitions.
2. Run the selected GLM, boosted, and pure-premium alternatives with saved configurations.
3. Evaluate once on the final holdout and generate tables/plots from a repeatable script.
4. Save full preprocessing-plus-model pipelines, feature lists, seeds, package versions, and target/reconciliation metadata.
5. Summarize findings against each research question; disclose uncertainty, limitations, and the CASdatasets source citation (Dutang and Charpentier).

Recommended minimum model set

Component	Minimum defensible model set
Frequency	Poisson GLM with exposure offset; one Poisson gradient-boosting model
Severity	Gamma log-link GLM; one nonlinear Gamma/severity model if supported
Pure premium	Frequency × severity; Tweedie GLM
Research extensions	Neural network and/or Tweedie boosting; include only if same-split validation and adequate reporting are feasible

> **Execution sequence:** reconcile → define targets → EDA → policy split → frequency GLM → severity GLM → boosted comparisons → Tweedie and frequency-severity premiums → tune on development data → test metrics/calibration → SHAP/group analysis → reproducible final report.

This is a recommended computational design based on the stated project questions and supplied dataset description. Final distributional assumptions and discrepancy handling must be justified from diagnostics, source documentation, and the research protocol.